Linear Programming in a Nutshell

Dominik Scheder

Abstract

This is a quick-and-dirty summary of Linear Programming, including most of what we covered in class. If you are looking for a readable, entertaining, and thorough treatment of Linear Programming, I can recommend "Understanding Linear Programming" by Gärtner and Matoušek [1].

In fact, this short note follows the that book in notation and concept, especially in the proof of the Farkas Lemma.

1 An Example

A farmer owns some land and considers to grow rice and potatoes and raise some cattle. On the market rice sells at 14 a unit, beef at 28, and potatoes at 7. Never mind which units of money, rice, and potatoes we have in mind. The three options also require different amounts of resources. For simplicity we assume that there are only two limited resources, namely land and water. The farmer has 35 units of land and can use 28 units of water each year. We summarize all data in one small table:

	Rice	Beef	Potato	Available
Land	1	8	1	35
Water	4	4	2	28
Market Price	14	28	7	

We see that producing one unit of rice requires 1 unit of land and 4 units of water. The farmer wants to maximize his annual revenue. Let r, b, p be variables describing the amount of rice, beef, and potatoes the farmer produces per year. We can write down the following maximizing problem:

$$
\begin{aligned}
\text{maximize} \quad & 14r + 28b + 7p \\
\\
\text{subject to} \quad & r + 8b + p \leq 35 \\
& 4r + 4b + 2p \leq 28 \\
\\
& r, b, p \geq 0 \ .
\end{aligned}
$$

1.1 Feasible Solutions

We see that every possible solution is a point $(r, b, p) \in \mathbb{R}^3$, but not every such point is possible. For example, $(10, 10, 10)$ would need too much water and too much land. We say this solution is *infeasible*. A *feasible* solution would for example be $(7, 0, 0)$, i.e., growing 7 units of rice and nothing else. This uses all the water and 7 of his 35 units of land. It yields and annual revenue of $14 \cdot 7$. This solution does not "feel" optimal, as most of the land stays unused. A better solution would be $(0, 7/2, 7)$, which uses all land and water, and yields a revenue of 147.

But wait: Can we grow three and a half cows? Surely cows only come in integers. What is half a cow supposed to be? Maybe a calf??? But remember, we talk about *units of beef*, and one unit is not necessarily one cow. It could be 10 *tons* of beef, for example. So $(0, 7/2, 7)$ is a feasible solution, it yields a revenue of 147. Furthermore, it uses up all resources, so it feels optimal. But it is not:

Consider $(3, 4, 0)$. This also uses all land and water but yields a revenue of 154. Is this optimal?

1.2 Upper Bounds, Economic Perspective

How can we derive upper bounds on the annual revenue the farmer can achieve? Here is an "economic" idea: Suppose a big agricultural firm approaches the farmer and offers him to rent out part of his land and water. The firm offers l units of money per unit of land, and w per unit of water. For example, suppose the firm offers $(l, w) = (0, 7)$. That means, it offers 7 per unit of water but 0 per unit of land. Accepting this offer, he would earn $35 \cdot 0 + 28 \cdot 7 = 196$ per year. But should he accept?

Let's see: Growing one unit of rice yields 14 on the market; but it uses 4 units of water, which he could rent to the firm at a total of 28. So no, growing rice makes no sense. How about beef? One unit yields 28 on the

market, but also uses 4 units of water, i.e., the farmer forgoes 28 units of money in rental income for every unit of beef. So raising cattle is not better than renting everything out to the firm. Potatoes: 7 units of money on the market, but uses 2 units of water, which would yield 8 units of money if rented out to the firm. So we see: The offer of the firm is at least as good as anything the farmer could grow by himself. We see that his annual income can never exceed 196 per year. Thus, 196 is an upper bound.

Let's put this into mathematical terms. Accepting an offer (l, w) by the firm would give him an annual income of $35l + 28w$. Under which conditions should he rent everything out to the firm? Suppose that

$$1 \cdot l + 4 \cdot w \geq 14 \ .$$

In this case the resources needed for growing one unit of rice could better be rented out to the firm (yielding $l + 4w$ of rental income) then grown as rice and sold on the market (yielding 14). For beef and potatoes we can write down similar inequalities. Thus, whenever (l, w) are such that

$$\begin{aligned} 1 \cdot l + 4 \cdot w &\geq 14 \\ 8 \cdot l + 4 \cdot w &\geq 28 \\ 1 \cdot l + 2 \cdot w &\geq 7 \ , \end{aligned}$$

then $35l + 28w$ is an upper bound on the farmer's income. Note that this is again an optimization problem:

$$\begin{array}{lrl} \text{minimize} & 35l + 28w & \\ & & \\ \text{subject to} & l + 4w & \geq 14 \\ & 8l + 4w & \geq 28 \\ & l + 2w & \geq 7 \\ & & \\ & w, l & \geq 0 \ . \end{array}$$

Note that w, l should be non-negative, as offering a negative price surely does not make sense. We have seen that $(0, 7)$ is a feasible solution of this minimization problem, yielding a value of 196. Another feasible solution is $(2, 3)$, yielding 154. Thus, the income of the farmer cannot exceed 154. But we have already seen that the farmer *can* achieve 154, by choosing $(r, b, p) = (3, 4, 0)$. Thus, we have proved that 154 is the optimal revenue the farmer can achieve.

1.3 Upper Bounds, Mathematical Perspective

Here is a different, less exciting, but more general view of the method by which we derived our upper bound. Let's again look at the original optimization problem:

$$\begin{array}{lrll} \text{maximize} & 14r + 28b + 7p & & \\ \\ \text{subject to} & r + 8b + p & \leq 35 & \text{(land)} \\ & 4r + 4b + 2p & \leq 28 & \text{(water)} \\ \\ & r, b, p & \geq 0 \ . & \end{array}$$

Let us multiply the first inequality (land) by l and the second (water) by w. If $l, w \geq 0$ we obtain

$$\begin{aligned} l(r + 8b + p) &\leq 35l \\ w(4r + 4b + 2p) &\leq 28w \end{aligned}$$

Note that l, w should not be negative, otherwise $\leq$ would become $\geq$. Adding up the two inequalities above, we obtain

$$\begin{aligned} l(r + 8b + p) + w(4r + 4b + 2p) &\leq 35l + 28w \iff \\ r(l + 4r) + b(8l + 4w) + p(l + 2w) &\leq 35l + 28w \ . \end{aligned}$$

We want this to be an upper bound on the farmer's revenue, i.e., on $14r + 28b + 7p$. That is, we want that

$$14r + 28b + 7p \leq r(l + 4r) + b(8l + 4w) + p(l + 2w) \ .$$

One way to achieve this would be to make sure that $14 \leq l+4r$, $28 \leq 8l+4w$, and $7 \leq l + 2w$. Thus, we want to choose (l, w) such that these three inequalities hold and $35l + 28w$ is as small as possible. We arrive at the following minimization problem:

$$\begin{array}{lrl} \text{minimize} & 35l + 28w & \\ \\ \text{subject to} & l + 4w & \geq 14 \\ & 8l + 4w & \geq 28 \\ & l + 4w & \geq 7 \\ \\ & w, l & \geq 0 \ . \end{array}$$

This is the same problem we derived through "economic reasoning".

2 Linear Programs

In our example the farmer had three variables to choose (r, b, p) and two constraints (land,water). Generally, a maximization problem of this form can have many variables $x_1, \ldots, x_n$ and several constraints. A *maximization linear program in standard form* is the following maximization problem:

$$\begin{array}{lll} \text{maximize} & c_1 x_1 + \cdots c_n x_n & \\ & & \\ \text{subject to} & a_{1,1}x_1 + \cdots + a_{1,n}x_n & \leq b_1 \\ & a_{2,1}x_1 + \cdots + a_{2,n}x_n & \leq b_2 \\ & \vdots & \\ & a_{m,1}x_1 + \cdots + a_{m,n}x_n & \leq b_m \\ & & \\ & x_1, \ldots, x_n & \geq 0 \ . \end{array}$$

This is a maximization linear problem with n variables and m constraints. We can write it succinctly in matrix-vector form:

$$\begin{array}{lll} & \text{maximize} & \mathbf{c}^T\mathbf{x} \\ P: & \text{subject to} & A\mathbf{x} \leq \mathbf{b} \\ & & \mathbf{x} \geq 0 \ . \end{array}$$

Here, $\mathbf{c} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, $A \in \mathbb{R}^{m \times n}$, i.e., A is a matrix of height m and width n. Finally, $\mathbf{x}$ is the column vector $(x_1, \ldots, x_n)^T$. A *feasible solution of* P is a vector $\mathbf{x} \in \mathbb{R}^n$ that satisfies the constraints, i.e., $A\mathbf{x} \leq \mathbf{b}$ and $\mathbf{x} \geq 0$. We denote the set of feasible solutions of P by $\text{sol}(P)$.

2.1 The value of a linear program

The value of P, or $\text{val}(P)$ for short, is the largest value $\mathbf{c}^T\mathbf{x}$ we can achieve for feasible solutions $\mathbf{x}$. So $\text{val}(P) := \max\{\mathbf{c}^T\mathbf{x} | \mathbf{x} \in \text{sol}(P)\}$. But wait: How do we know that this set has a maximum? First, it could be empty: What if P contains contradicting constraints? Second, it could contain arbitrarily large numbers. Even worse, it might be non-closed, like $[0, 1)$. So let us be more careful with our definition of $\text{val}(P)$.

Definition 1 (The Value of a Maximization Linear Program)**.** *Let P be a maximization linear program. Let $S := \{\mathbf{c}^T\mathbf{x} | \mathbf{x} \in \text{sol}(P)\}$. Then*

$$\text{val}(P) := \begin{cases} -\infty & \text{if } S = \emptyset \ . \\ +\infty & \text{if } S \text{ contains arbitrarily large numbers,} \\ \sup(S) & \text{else.} \end{cases}$$

Furthermore, if $S =$ we say P is infeasible. *In the second case we say it is* unbounded, *and in the third case we say it is* feasible and bounded.

2.2 The Dual of a Linear Program

Just as we did with our "economic" or "mathematical" reasoning, we can derive upper bounds on $\text{val}(P)$ by multiplying each constraint by a non-negative number y_i and summing up. This leads to the following *minimization linear program*:

$$\begin{array}{lll} & \text{minimize} & \mathbf{b}^T\mathbf{y} \\ D: & \text{subject to} & A^T\mathbf{y} \geq \mathbf{c} \\ & & \mathbf{x} \geq 0 \ . \end{array}$$

This is a program with m variables and n constraints. We call D the *dual program of P*. In analogy to $\text{val}(P)$, we define the value of a minimization problem as follows: Let $T := \{\mathbf{b}^T\mathbf{y} | \mathbf{y} \in \text{sol}(D)\}$. Then

$$\text{val}(D) := \begin{cases} +\infty & \text{if } T = \emptyset \ , \\ -\infty & \text{if } T \text{ contains arbitrarily small numbers,} \\ \inf(S) & \text{else.} \end{cases}$$

2.3 Weak Duality

Theorem 2. *Let P be a maximization linear program and D its dual. If $\mathbf{x} \in \text{sol}(P)$ and $\mathbf{y} \in \text{sol}(D)$, then $\mathbf{c}^T\mathbf{x} \leq \mathbf{b}^T\mathbf{y}$.*

If the reader has understood how we derived the minimization problem in the farming example, the proof should already be evident. Still, let us give a formal three-line proof.

Proof. Since $\mathbf{y}$ is feasible for D, we have $\mathbf{c}^T \leq (A^T\mathbf{y})^T = \mathbf{y}^T A$. Since $\mathbf{x} \geq 0$ this means $\mathbf{c}^T\mathbf{x} \leq \mathbf{y}^T A\mathbf{x}$. Since $\mathbf{x}$ is feasible for P we know $A\mathbf{x} \leq \mathbf{b}$, and since $y^T \geq 0$ this implies $\mathbf{y}^T A\mathbf{x} \leq \mathbf{y}^T\mathbf{b} = \mathbf{b}^T\mathbf{y}$. □

This theorem has an immediate corollary:

Theorem 3 (Weak LP Duality Theorem)**.** *Let P be a maximization LP and D its dual. Then* $\mathrm{val}(P) \leq \mathrm{val}(D)$.

Proof. This follows from the previous theorem plus some case distinction about whether P, D are unbounded, infeasible, etc. □

2.4 Linear Programs in General Form

First note that a maximization LP can easily be transformed into an equivalent minimization problem, namely

$$\begin{array}{lll} & \text{minimize} & (-\mathbf{c}^T)\mathbf{x} \\ -P: & \text{subject to} & (-A)\mathbf{x} \geq -\mathbf{b} \\ & & \mathbf{x} \geq 0\ . \end{array}$$

It should be clear that $\mathrm{sol}(-P) = \mathrm{sol}(P)$ and $\mathrm{val}(-P) = -\mathrm{val}(P)$. So there is really no significant difference between maximization and minimization problems.

Second, note that naturally an optimization problem could come with some *equality constraints* and some *unbounded variables*. One example of such a *linear program in general form* is

$$\begin{array}{lll} & \text{maximize} & 2e + f + g \\ & & \\ & \text{subject to} & e \leq 3 \qquad (a) \\ & & e + f \leq 1 \qquad (b) \\ P: & & f + g = 1 \qquad (c) \\ & & g \leq 2 \qquad (d) \\ & & \\ & & e, g \geq 0, f \in \mathbb{R}\ . \end{array}$$

To make dualization easier, it is a good idea to assign names to our constraints from the very beginning, like (a)–(d) in the above case. Can we transform P into an equivalent LP P' in standard form? The equality constraint (c) : $f + g = 1$ is easily taken care of: We can replace it by $f + g \leq 1$ and $-f - g \leq -1$. The variable $f \in \mathbb{R}$ is more challenging: We introduce two new variables $f_1, f_2 \geq 0$ and replace every occurrence of f by $f_1 - f_2$. We

obtain:

$$
\begin{array}{lll}
 & \text{maximize} & 2e + f_1 - f_2 + g \\
 & & \\
 & \text{subject to} & \begin{array}{rll} e & \leq 3 & (a) \\ e + f_1 - f_2 & \leq 1 & (b) \\ f_1 - f_2 + g & \leq 1 & (c_1) \\ -f_1 + f_2 - g & \leq -1 & (c_2) \\ g & \leq 2 & (d) \end{array} \\
 & & \\
 & & e, f_1, f_2, g \geq 0 \ .
\end{array}
\qquad P' :
$$

The reader should check that these two programs have indeed the same value. What is the dual D' of P'? It has five variables a, b, c_1, c_2, d and four constraints $(e), (f_1), (f_2), (g)$:

$$
\begin{array}{lll}
 & \text{minimize} & 3a + b + c_1 - c_2 + 2d \\
 & & \\
 & \text{subject to} & \begin{array}{rll} a + b & \geq 2 & (e) \\ b + c_1 - c_2 & \geq 1 & (f_1) \\ -b - c_1 + c_2 & \geq -1 & (f_2) \\ c_1 - c_2 + d & \geq 1 & (g) \end{array} \\
 & & \\
 & & a, b, c_1, c_2, d \geq 0 \ .
\end{array}
\qquad D' :
$$

Note that the constraints (f_1) and (f_2) can be combined into one equality constraint $(f) : b + c_1 - c_2 = 1$. Furthermore, c_1 and c_2 as the pair $(c_1 - c_2)$. This difference ranges over all $\mathbb{R}$ and thus we can replace it by $c \in \mathbb{R}$. We finally arrive at the dual D in general form:

$$
\begin{array}{lll}
 & \text{minimize} & 3a + b + c + 2d \\
 & & \\
 & \text{subject to} & \begin{array}{rll} a + b & \geq 2 & (e) \\ b + c & = 1 & (f) \\ c + d & \geq 1 & (g) \end{array} \\
 & & \\
 & & a, b, d \geq 0, \ c \in \mathbb{R} \ .
\end{array}
\qquad D' :
$$

We observe: The primal equality constraint $(c) : f + g = 1$ translates into an unbounded dual variable $c \in \mathbb{R}$, and the unbounded primal variable f translates into a dual equality constraint $(f) : b + c = 1$.

The reader should check that these rules hold in general. Also, they can be derived by more general reasoning: If we have a primal inequality of the form $\cdots \leq \ldots$, we must make sure that the multiplier y_i is non-negative, otherwise $\leq$ becomes $\geq$. However, if it is an equality constraint $\cdots = \ldots$, we can multiply it with a negative value y_i, too. Also, for deriving upper bounds we argued that $\mathbf{c}^T\mathbf{x} \leq (\mathbf{y}^T A)\mathbf{x}$ we used the fact $\mathbf{x} \geq 0$. However, if some $x_j \in \mathbb{R}$ is unbounded, this is not valid anymore, unless the jth coordinate of $\mathbf{c}^T$ *equals* the jth coordinate of $(\mathbf{y}^T A)$.

3 Existence of Optimal Solutions

Suppose P is feasible and bounded. Then we defined $\text{val}(P) := \sup\{\mathbf{c}^T\mathbf{x} | \mathbf{x} \in \text{sol}(P)\}$. In this section we prove that this supremum is indeed a maximum:

Theorem 4 (Existence of Optimal Solutions)**.** *Suppose P is feasible and bounded. Then there is some $\mathbf{x}^* \in \text{sol}(P)$ such that $\mathbf{c}^T\mathbf{x} \leq \mathbf{c}^T\mathbf{x}^*$ for all $\mathbf{x} \in \text{sol}(P)$.*

It turns out that this theorem is marginally easier to prove if we assume that P is a minimization problem:

$$\begin{array}{lll} & \text{minimize} & \mathbf{c}^T\mathbf{y} \\ P: & \text{subject to} & A\mathbf{x} \geq \mathbf{b} \\ & & \mathbf{x} \geq 0\ . \end{array}$$

Here $A \in \mathbb{R}^{m \times n}$, so we have n variables and m constraints. Note that we really have $m + n$ constraints, since $x_j \geq 0$ is a constraint, too! Thus let us write P in even more compact form

$$\begin{array}{lll} & \text{minimize} & \mathbf{c}^T\mathbf{y} \\ P: & \text{subject to} & A\mathbf{x} \geq \mathbf{b} \end{array}$$

and keep in mind that the n last rows of A form the identity matrix I_n. We introduce some notation: $\mathbf{a}_i$ is the ith row of i; for $I \subseteq [m+n]$ let A_I be the matrix consisting of the rows $\mathbf{a}_i$ for $i \in I$.

Definition 5. *For $x \in \mathbb{R}^n$ let $I(x) := \{i \in [m+n] \mid \mathbf{a}_i\mathbf{x} = b_i\}$ be the set of indices of the constraints that are "tight", i.e., satisfied with equality. We call $\mathbf{x} \in \mathbb{R}^n$ a* basic point *if $\text{rank}(A_{I(x)}) = n$. If $\mathbf{x}$ is a basic point and feasible, we call it a* basic feasible solution *or simply a* basic solution*.*

Proposition 6. *The program P has at most $\binom{m+n}{n}$ basic points.*

Proof. Let $\mathbf{x}$ be a basic point. Thus $\text{rank}(A_{I(\mathbf{x})}) = n$. By basic linear algebra there must be a set $I \subseteq I(\mathbf{x})$ such that $|I| = n$ and $\text{rank}(A_I) = n$. Furthermore, such a set uniquely specifies $\mathbf{x}$ since $A_I\mathbf{x} = \mathbf{b}_I$ has exactly one solution. There are at most $\binom{m+n}{n}$ sets I of size n such that A_I has rank n, and therefore there are at most that many basic points. □

Lemma 7. *Suppose P is bounded and $\mathbf{x} \in \text{sol}(P)$ is a feasible solution. Then there exists a basic feasible solution $\mathbf{x}^*$ which is at least as good as $\mathbf{x}$, namely $\mathbf{c}^T\mathbf{x}^* \leq \mathbf{c}^T\mathbf{x}$ (recall that P is a minimization problem).*

Proof of Theorem 4. : If P is feasible, then by the above lemma, it has at least one basic feasible solution. Let $\mathbf{x}^*$ be the basic feasible solution that maximizes $\mathbf{c}^T\mathbf{x}^*$ among all basic feasible solutions. This exists, as there are only finitely many. Again by the lemma, this must be an optimal solution. □

Proof of the lemma. Let $\mathbf{x}$ be a feasible solution that is not basic. We will move $\mathbf{x}$ along some line $\mathbf{x}^t := \mathbf{x} + t\mathbf{z}$ such that (i) $\mathbf{x}^t$ stays feasible; (ii) the value $\mathbf{c}^T x^t$ does not increase; (iii) at some point one more constraint becomes tight, i.e., the size of $I(\mathbf{x}^t)$ increases. Repeating this process will terminate, as $|I(\mathbf{x}^t)|$ cannot exceed $n + m$.

Let's elaborate on the details. Set $I := I(\mathbf{x})$ for brevity. The point $\mathbf{x}$ is not basic, which means $\text{rank}(A_I) < n$. Thus we can find a vector $\mathbf{z} \in \mathbb{R}^n$ such that $A_I\mathbf{z} = 0$. Let $J := [m + n] \setminus I$. We have:

$$A_I\mathbf{x} = b_I$$
$$A_J\mathbf{x} > b_J \ .$$

Set $\mathbf{x}^t := \mathbf{x} + t\mathbf{z}$. Then $A_I\mathbf{x}^t = A_I\mathbf{x} + tA_I\mathbf{z} = b_I$ and $A_J\mathbf{x}^t = A_J\mathbf{x} + tA_J\mathbf{z}$. This means that $\mathbf{x}^t$ is feasible as long as $|t|$ is sufficiently small.

Case 1. $\mathbf{c}^T\mathbf{z} < 0$. If we start with $t = 0$ and slowly increase t, our target function $\mathbf{c}^T\mathbf{x}^t$ decreases (which is good, as we want to minimize it). Now two things can happen: For some $t > 0$ one of the constraints in $A_J\mathbf{x}^t \geq \mathbf{b}_J$ becomes tight, i.e., $\mathbf{a}_i\mathbf{x} + t\mathbf{a}_i\mathbf{z} = b_i$ for some i. In this case we are done, since $I(\mathbf{x}^t)$ has now grown by at least 1. If this doe not happen, i.e., no additional constraint becomes tight, for any $t \geq 0$, then all $\mathbf{x}^t$ are feasible and P is clearly unbounded, contradicting our assumption.

Case 2. $\mathbf{c}^T\mathbf{z} > 0$. This is analogous to the previous case, just that we start with $t = 0$ and slowly decrease it.

Case 3. $\mathbf{c}^T\mathbf{z} = 0$. Note that we can assume $\mathbf{z}_j < 0$ for some j, since otherwise we can replace $\mathbf{z}$ by $-\mathbf{z}$, which is also in $\ker(A_I)$. Note th

Now we start at $t = 0$ and slowly increase t until some constraint of $A_J\mathbf{x} \geq \mathbf{b}$ becomes tight. This must happen: Increasing t decreases x_j^t and it will finally become 0, making the constraint $x_j \geq 0$ tight.

We see that in all three cases the number of tight constraints increases (unless P is unbounded). Thus we iterate this process and will eventually terminate with a basic solution. □

4 Farkas Lemma

In order to prove the strong duality theorem we have to take a brief detour.

Lemma 8 (Farkas Lemma)**.** *Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$. Then exactly one of the following two statements hold.*

1. *There exists some $\mathbf{x} \in \mathbb{R}^n$ such that $A\mathbf{x} \leq \mathbf{b}$.*
2. *There exists some $\mathbf{y} \in \mathbb{R}^m$, $y \geq 0$, such that $\mathbf{y}^T A = 0$ and $\mathbf{y}^T b < 0$.*

Proof. First we show that the two statements cannot both be true. Suppose Point 1 holds. Consider any $y \in \mathbb{R}^m$ with $\mathbf{y} \geq 0$. Since $A\mathbf{x} \leq \mathbf{b}$ and $\mathbf{y} \geq 0$ it holds that $\mathbf{y}^T A\mathbf{x} \leq \mathbf{y}^b$. So it cannot be that the left-hand side is 0 and the right-hand side is negative. So Point 2 cannot hold.

Next we have to show that at least one of these statements is true. We do so by induction on n. The base case $n = 0$ is easy but weird and left for the reader as an exercise.

Suppose $n \geq 1$ and let us denote the system $A\mathbf{x} \leq \mathbf{b}$ by P. P has m inequalities. We partition $[m] = C \cup F \cup L$ as follows: Consider an inequality $a_i\mathbf{x} \leq b_i$, and in particular $a_{i,1}$, the coefficient of x_1. If $a_{i,1} > 1$, this inequality provides an upper bound (ceiling) for x_1, and we put i into C. If $a_{i,1} < 1$ it is a lower bound (floor) for x_1 and we put i into F. If $a_{i,1} = 0$ it provides no bound on x_1 and we put x_1 into L (level). By multiplying our rows by an appropriate (positive) constant, we can make sure that $a_{i,1} \in \{-1, 0, 1\}$ for all $i \in [m]$. This gives the following equivalent system with variables

$\mathbf{x}' = (x_2, \ldots, x_n)$:

$$\begin{aligned} x_1 + a'_i\mathbf{x}' &\leq b'_i \ \forall i \in C \\ -x_1 + a'_j\mathbf{x}' &\leq b'_j \ \forall j \in F \\ a'_k\mathbf{x}' &\leq b'_k \ \forall k \in L \ . \end{aligned} \tag{P'}$$

Note that his is of the form $A'\mathbf{x} \leq \mathbf{b}'$ where $A' = T_1A$, $\mathbf{b}' = T_1A$ for some matrix T_1 that encodes our multiplication by positive numbers. So $T_1 \geq 0$, meaning every entry is non-negative.

Note that if we add some inequality in C and some in F, the variable disappears. Doing this for all $|C| \ldots |F|$ pairs, we get the following system:

$$\begin{aligned} (a'_i + a'_j)\mathbf{x}' &\leq b'_i + b'_j \ \forall i \in C, j \in F \ , \\ a'_k\mathbf{x}' &\leq b'_k \ . \end{aligned}$$

This is a system of $|C| \cdot |F| + |L|$ inequalities over $n-1$ variables and can be succinctly written as

$$\bar{A}\mathbf{x}' \leq \bar{b} \tag{Q}$$

Note that the system arises from P' by adding up certain inequalities. So $\bar{A} = T_2A'$, $\bar{b} = T_2\mathbf{b}'$ where again $T_2 \geq 0$. In fact, it is easy to see that every entry of T_2 is either 0 or 1. Thus we see that $\bar{A} = TA, \bar{\mathbf{b}} = T\mathbf{b}$, for some matrix $T \geq 0$. We have the following proposition:

Proposition 9. *Let P, Q as above. If $(x_1, \ldots, x_n)$ is a feasible solution of P, then $(x_2, \ldots, x_n)$ is a feasible solution of Q. Conversely, if $(x_2, \ldots, x_n)$ a feasible solution of Q, then there exists some $x_1 \in \mathbb{R}$ such that $(x_1, \ldots, x_n)$ is a feasible solution of P.*

Furthermore, there is a matrix $T \geq 0$ such that $\bar{A} = TA$ and $\bar{b} = T\mathbf{b}$.

Proof. The first statement should be obvious. So suppose $\mathbf{x}' = (x_2, \ldots, x_n)$ satisfies Q. Since $(a'_i + a'_j)\mathbf{x}' \leq b'_i + b'_j$ for all $i \in C, j \in F$, we see that

$$\begin{aligned} a'_j\mathbf{x}' - b'_j \leq b'_i - a'_i\mathbf{x}' \quad \forall i \in C, j \in F \ , \quad \text{and therefore} \\ \max_{j \in F} a'_j\mathbf{x}' - b'_j \leq \min_{i \in C} b'_i - a'_i\mathbf{x}' \ . \end{aligned}$$

Choose some value x_1 between the max and the min. Then $-x_1 + a'_j\mathbf{x}' \leq b'_j$ and $x_1 + a'_i\mathbf{x}' \leq b'_i$. Furthermore $a'_k\mathbf{x}' \leq b'_k$ holds anyway since $\mathbf{x}'$ satisfies Q. Thus we see that $\mathbf{x} := (x_1, \ldots, x_n)$ satisfies P' and therefore P. □

We can finish the proof Farkas Lemma. If $P : A\mathbf{x} \leq \mathbf{b}$ is feasible, we are done. Otherwise, it is infeasible is so is P'. By induction on n we know that there exists some vector $\bar{y} \in \mathbb{R}^{|C|\cdot|F|+|R|}$, $\bar{y} \geq 0$ such that $\bar{y}^T A = \mathbf{0}$ and $\bar{y}^T \bar{b} < 0$. We set $\mathbf{y} := \bar{y}T$. Since $T \geq 0$ also $\mathbf{y} \geq 0$ and

$$\begin{aligned} \mathbf{y}^T A &= \bar{y}^T T A = \bar{y}^T A = 0 \\ \mathbf{y}^T \mathbf{b} &= \bar{y}^T T \mathbf{b} = \bar{y}^T \bar{b} < 0 \ . \end{aligned}$$

Thus the vector $\mathbf{y}$ satisfies the condition of Point 2, which finishes the proof. □

5 Strong LP Duality

First we need a different version of Farkas Lemma:

Lemma 10 (Farkas Lemma, Version 2)**.** *Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$. Then exactly one of the following two statements hold.*

1. *There exists some $\mathbf{x} \in \mathbb{R}^n$, $x \geq 0$ such that $A\mathbf{x} \leq \mathbf{b}$.*
2. *There exists some $\mathbf{y} \in \mathbb{R}^m$, $y \geq 0$, such that $\mathbf{y}^T A \geq 0$ and $\mathbf{y}^T b < 0$.*

The reader can verify that the first version of Farkas Lemma implies the second.

Let us now again consider a linear program P and its dual D:

$$\begin{array}{lll} & \text{maximize} & \mathbf{c}^T\mathbf{x} \\ P: & \text{subject to} & A\mathbf{x} \leq \mathbf{b} \\ & & \mathbf{x} \geq 0 \ . \end{array}$$

$$\begin{array}{lll} & \text{minimize} & \mathbf{b}^T\mathbf{y} \\ D: & \text{subject to} & A^T\mathbf{y} \geq \mathbf{c} \\ & & \mathbf{x} \geq 0 \ . \end{array}$$

5.1 Strong LP Duality, Warm-Up

As a warm-up we show the following theorem:

Theorem 11. *Suppose P is infeasible. Then D is either infeasible or unbounded.*

Proof. Suppose P is infeasible and D is feasible. We have to show that it is unbounded. First of all, there is a dual feasible solution $\mathbf{y}$. Second, the system

$$\begin{aligned} A\mathbf{x} &\leq \mathbf{b} \\ \mathbf{x} &\geq 0 \end{aligned}$$

has no solution, since P is infeasible. By Farkas Lemma Version 2 there is some $z \in \mathbb{R}^m$ such that $\mathbf{z} \geq 0$, $\mathbf{z}^T A \geq 0$ and $\mathbf{z}^T \mathbf{b} < 0$. Now consider the point

$$\mathbf{y}(t) := \mathbf{y} + t\mathbf{z} \ .$$

We claim that $\mathbf{y}(t)$ is feasible for D for every $t \geq 0$: Clearly $\mathbf{y} + t\mathbf{z} \geq 0$ since $\mathbf{y}, \mathbf{z}, t \geq 0$. Also $A^T \mathbf{y}^T = A^T(\mathbf{y} + t\mathbf{z}) = A^T\mathbf{y} + tA^T\mathbf{z} \geq 0$ by assumption on $\mathbf{y}, \mathbf{z}$. So the claim is proved.

Finally observe that he value of D under solution $\mathbf{y}(t)$ is $\mathbf{b}^T\mathbf{y} = \mathbf{b}^T\mathbf{y} + t\mathbf{b}^T\mathbf{z}$. Since $\mathbf{b}^T\mathbf{z} < 0$ and t can be made arbitrarily large, the value is $-\infty$, i.e., D is unbounded. □

With a similar proof we get

Theorem 12. *Suppose D is infeasible. Then P is either infeasible or unbounded.*

5.2 The Real Thing: Strong LP Duality

Theorem 13. *Suppose P is feasible and bounded. Then D is feasible and bounded, too, and* $\mathrm{val}(P) = \mathrm{val}(D)$.

Proof. If D is unbounded, P must be infeasible by weak duality. Thus, we can conclude that D is not unbounded. If D is infeasible, then by Theorem 12 P is either infeasible or unbounded. So D cannot be infeasible. We conclude that D is feasible and bounded, also.

Having settled that both P, D are feasible and bounded, let $\alpha := \mathrm{val}(P)$, $\beta := \mathrm{val}(D)$. By weak duality we know that $\alpha \leq \beta$. We want to prove that they are in fact equal. So suppose, for the sake of contradiction, that $\alpha < \beta$. For $\gamma \in \mathbb{R}$ define the following system of inequalities P_γ:

$$\begin{aligned} \mathbf{c}^T\mathbf{x} &\geq \gamma \\ A\mathbf{x} &\leq \mathbf{b} \\ \mathbf{x} &\geq \mathbf{0} \ . \end{aligned}$$

Note that P_α is satisfiable (by the optimal solution of P, for example) but P_β is not (otherwise the value of P would be at least β). We can bring P_β in matrix-vector form:

$$\left[\frac{-\mathbf{c}^T}{A}\right]\mathbf{x} \leq \left[\frac{-\beta}{\mathbf{b}}\right]$$

This system has no solution (since $\text{val}(P) < \beta$) and thus there exists a row vector $\bar{\mathbf{y}}^T = [z|\mathbf{y}^T] \in \mathbb{R}^{m+1}$, non-negative, such that

$$[z|\mathbf{y}^T] \cdot \left[\frac{-\mathbf{c}^T}{A}\right] \geq \mathbf{0}\ , \qquad [z|\mathbf{y}^T] \cdot \left[\frac{-\beta}{\mathbf{b}}\right] < 0\ .$$

Equivalently:

$$\mathbf{y}^T A \geq z\mathbf{c}^T$$
$$\mathbf{y}^T\mathbf{b} < z\beta\ .$$

Suppose $z > 0$. Then set $\mathbf{v} := \frac{1}{z}\mathbf{y}$ and observe that $\mathbf{v} \geq 0$, $A^T v \geq \mathbf{c}$ and $\mathbf{b}^T\mathbf{v} < \beta$. In other words, $\mathbf{v}$ is a feasible solution for D and gives a value *less than* β, which is a contradiction.

If z is not positive, it must be 0 (its non-negativity is guaranteed by Farkas Lemma Version 2). So suppose it is 0. Then $\mathbf{y}$ satisfies the following two inequalities:

$$\mathbf{y}^T A \geq \mathbf{0} \tag{1}$$
$$\mathbf{y}^T\mathbf{b} < 0\ . \tag{2}$$

We will soon derive a contradiction. Let x be a feasible solution of P, i.e., $\mathbf{x} \geq 0$ and $A\mathbf{x} \leq \mathbf{b}$. But then

$$\begin{aligned} 0 &\leq \mathbf{y}^T A\mathbf{x} && \text{(by (1) and } \mathbf{x} \geq \mathbf{0}\text{)} \\ &\leq \mathbf{y}^T\mathbf{b} && \text{(by } \mathbf{y} \geq \mathbf{0} \text{ and } A\mathbf{x} \leq \mathbf{b}\text{)} \\ &< 0\ . && \text{(by (2))} \end{aligned}$$

This is clearly a contradiction.

We have shown how the assumption $\alpha < \beta$ yields a vector $[z|\mathbf{y}^T] \geq 0$, which gives a contradiction both for $z > 0$ and $z = 0$. The theorem is proved. □

Putting everything together, we arrive at the following theorem:

Theorem 14. *Let P be a linear maximization program and D its dual. Then* $\text{val}(P) = \text{val}(D)$ *unless both are infeasible.*

References

[1] Bernd Gärtner and Jiří Matoušek. *Understanding and Using Linear Programming*. Universitext. Springer, 1 edition, 2006.